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VeriSight

AI-Powered Visual Verification System

for Delivery Platform Fraud Detection

Team	Domain
return0	**Data Science, AI & Advanced Computing**

Core Tech	Latency Target	Problem Scale
CNN | ViT | GAN | OCR	**< 5 seconds end-to-end**	**USD 103 Billion / year**

1. Executive Summary

E-commerce, quick commerce, and food delivery platforms across India and the world process millions of refund and complaint requests every day. The vast majority of these claims rely entirely on user-uploaded product images as visual proof that something went wrong with their order.

This trust-based system is being systematically exploited. Customers are using freely available AI image editing tools, generative AI platforms, and digital editors to manipulate product images before submitting them as complaint evidence. The most common targets are product expiry dates — altered to make a fresh product appear expired — and packaging labels, which are digitally modified to create false impressions of damage, wrong contents, or contamination.

> **Key Statistic**
> Returns fraud and abuse accounted for 13.7% of all returns in 2023, amounting to over USD 100 billion in losses globally. In India, abusive returns soared 64% between January 2024 and May 2025.

VeriSight is an AI-powered visual verification system purpose-built to detect these image manipulations in real time. It combines CNN-based image forensics, Transformer-based AI-generated image detection, GAN artifact analysis, and OCR-assisted text localization to evaluate every complaint image and produce an Authenticity Confidence Score — a single, explainable metric that tells the platform whether a submitted image is genuine or manipulated.

VeriSight is designed as a cloud-ready, API-first microservice that integrates directly into the complaint and refund workflows of any delivery-based platform — Zomato, Swiggy, Blinkit, BigBasket, Amazon, Flipkart, and beyond. It is practical, scalable, and deployable — not just a research prototype.

2. Problem Statement

2.1 The Refund Complaint Ecosystem

Modern delivery platforms have built customer trust on generous and fast refund policies. When a customer claims they received an expired product, wrong item, damaged packaging, or contaminated food, platforms typically process the refund with minimal verification — relying almost entirely on the image submitted by the customer as proof.

This trust-first approach, while excellent for customer experience, has created a massive and growing attack surface. Fraudsters have discovered that submitting a manipulated image is the easiest path to an unauthorized refund — and with AI tools now freely accessible to anyone with a smartphone, image manipulation has become trivially easy.

2.2 Core Fraud Patterns VeriSight Targets

Fraud Type	Description
Expiry Date Manipulation	Customer digitally alters the printed expiry date on packaging — changing a valid future date to a past date — to falsely claim the delivered product was expired at the time of delivery.
Packaging Label Manipulation	Brand name, product name, batch number, MRP, weight, or ingredient list on packaging is digitally edited to create the appearance of wrong or counterfeit product delivery.
AI Image Generation	Customer uses generative AI tools (Stable Diffusion, DALL-E, Adobe Firefly, Midjourney) to create an entirely synthetic product complaint image that never existed in reality.
GAN-Manipulated Images	GAN-based inpainting tools realistically alter specific regions of a product photo — inserting damage, mold, incorrect labeling, or wrong packaging — while keeping the rest photorealistic.
Image Splicing	Content from a different product image is composited into the complaint photo — for example, a damaged portion of another product's packaging is pasted over the delivered product's intact packaging.
Recycled Complaint Images	Previously used complaint images (genuine or fake) are resubmitted for new orders across the same or multiple accounts to claim repeated refunds without submitting new manipulated images.

2.3 Why Existing Systems Fail

- Platforms rely on human reviewers who cannot detect AI-generated or subtly edited images with the naked eye.
- No automated system exists that specifically verifies the authenticity of complaint product images.
- Manual review is slow (24–48 hours), inconsistent across reviewers, and completely unscalable at platform volumes.
- Current rule-based fraud filters (e.g., flag if same user complains 3 times) do not examine image authenticity at all.
- Generative AI tools produce images with photorealistic quality indistinguishable from real photos without algorithmic analysis.

- Platforms that do have image checks typically use simple hash-based duplicate detection — easily bypassed by minor edits.

2.4 Market Scale & Financial Impact

Metric	Data
Global e-commerce fraud losses (2024)	USD 48 billion annually. Businesses lose an average of USD 207 for every USD 100 in fraud when factoring in chargebacks, shipping, and processing costs.
Return & refund fraud globally (2024)	USD 103 billion. Returns fraud accounted for 13.7% of all returns in 2023. Abusive returns soared 64% between January 2024 and May 2025.
India deepfake shopping scams (2024)	45% of Indian consumers or someone they know fell victim to deepfake shopping scams (McAfee Global Festive Shopping Survey 2024).
India example: Meesho fraud	A single organized group in Surat defrauded Meesho of Rs 5.5 crore through coordinated refund fraud — illustrating the organized scale of the problem.
India consumer complaints growth	7,482 consumer grievances filed against food delivery apps in 2024–25 alone — a 10x increase from 805 in 2020–21.

3. Solution Overview — The DS/AI Approach

VeriSight is an AI-powered visual verification system that automatically analyzes every user-submitted complaint image and produces an Authenticity Confidence Score between 0 and 100, along with a human-readable explanation of the factors that contributed to the score. The system is built around four core analysis modules that work in parallel, each targeting a distinct manipulation vector.

> **VeriSight Core Philosophy**
> VeriSight does NOT make the refund decision — it provides an evidence-based Authenticity Confidence Score with explainable insights that enables the platform to make a faster, fairer, and more accurate decision. The system supports human reviewers rather than replacing them.

3.1 Why This is a DS/AI & Advanced Computing Problem

- Multimodal Deep Learning: Simultaneously processing visual features (CNN/ViT), frequency-domain signals (FFT/DWT), semantic embeddings (CLIP), and text extraction (OCR) — a genuinely multimodal AI inference problem.
- Adversarial AI Detection: VeriSight is an AI system built to detect and counter other AI systems (GANs, diffusion models, generative AI tools) — a core advanced computing challenge at the frontier of AI safety.
- Explainable AI (XAI): SHAP-based feature attribution and Grad-CAM activation maps make every fraud decision interpretable and legally defensible — applied XAI for high-stakes real-world decisions.
- Distributed ML Inference: Celery-based parallel task orchestration runs four AI models concurrently, achieving sub-5-second end-to-end latency at scale — advanced computing systems design.
- Custom Dataset Construction: VeriSight builds a novel labeled dataset of manipulated product packaging images (4,000+ samples, 5 classes) — a data science contribution with no existing public analog.

4. Core Detection Modules

4.1 Module 1 — CNN-Based Image Forensics (Weight: 35%)

Purpose

A deep Convolutional Neural Network forms the foundational analysis layer of VeriSight. It performs pixel-level analysis of the submitted complaint image to detect signs of manual editing, region-level manipulation, and image compositing.

What It Detects

- Copy-move forgery: Areas of the image that have been duplicated and repositioned.
- Image splicing: Seamless insertion of content from a different image source, detectable through pixel-level statistical inconsistencies at boundary regions.
- Error Level Analysis (ELA): Edited JPEG images show inconsistent compression artifact levels across regions; ELA maps reveal these as visually distinct areas.
- Lighting and shadow inconsistencies: Digitally inserted regions typically have lighting angles and shadow directions that do not match the rest of the image.
- Noise pattern analysis: Genuine images have consistent sensor noise patterns; inserted regions from different sources show different noise profiles.

Model Architecture

Component	Details
Primary Backbone	EfficientNet-B4 pre-trained on ImageNet — 7.6x smaller and 5.7x faster than ResNet-152 with higher accuracy on forgery benchmarks.
Forgery Localization	GCA-Net (Gated Context Attention Network) for pixel-level manipulation region segmentation — produces visual heatmaps showing exactly which pixels were manipulated.
ELA Pipeline	Custom ELA-CNN pipeline that generates Error Level Analysis maps and feeds them as additional input channels alongside the RGB image to the classification model.
Training Dataset	CASIA 2.0 (12,614 images), CoMoFoD (copy-move forgery), MICC-F220 (forgery localization ground truth), DF2023 (1 million+ images).
Target Accuracy	>95% on CASIA 2.0 using EfficientNet-B4 + ELA pipeline combination (ResNet-101 baseline: 93.46%, IEEE 2024).

4.2 Module 2 — Vision Transformer AI-Generated Image Detection (Weight: 30%)

Purpose

This module specifically targets complaint images that were entirely or partially generated using modern generative AI tools — including Stable Diffusion, DALL-E 3, Midjourney, Adobe Firefly, and GAN-based image generators.

Why Transformers for This Task

Vision Transformers (ViTs) use self-attention mechanisms to model relationships between non-adjacent image patches — enabling them to detect long-range semantic inconsistencies that CNNs miss. Research shows ViTs achieve up to 96% accuracy on AI-generated image detection tasks. Crucially, transformers generalize better to unseen generative models than CNN-only approaches.

Detection Mechanisms

- Semantic inconsistency detection: AI-generated product images often contain logically inconsistent text, labels, or barcodes (e.g., partially hallucinated expiry dates, wrong brand fonts).
- Texture artifact analysis: GAN and diffusion model generated images often produce overly smooth textures or unnatural repetitive patterns in background regions.
- Frequency domain analysis: AI-generated images have characteristic spectral signatures in the frequency domain (detectable via Discrete Fourier Transform and Discrete Wavelet Transform).
- CLIP-based feature analysis: CLIP embeddings (vision-language model) inherently distinguish between AI-generated and real images; CLIP-ResNet50 achieves 99.4% accuracy on StyleGAN detection (CVPR 2023).

Model Stack

Model	Role & Benchmark
ViT-B/16	Primary AI-generated image classifier. Achieves up to 96% accuracy using self-attention mechanisms to detect inter-patch semantic inconsistencies.
CLIP-ResNet50	GAN fingerprint detector. Achieves 99.4% and 99.2% accuracy on StyleGAN and StyleGAN2 generated images respectively (CVPR 2023, LGrad framework).
FatFormer	Generalizable synthetic image detection integrating Forgery-Aware Adapter and Language-Guided Alignment — effective on unseen generative models including diffusion models.
Grad-CAM XAI	Gradient Class Activation Mapping generates visual heatmaps of which image regions triggered the AI-detection decision — essential for human reviewer understanding.

4.3 Module 3 — GAN Artifact & Texture Inconsistency Analysis (Weight: 20%)

Purpose

Even when a complaint image is not fully AI-generated, GAN-based inpainting tools can be used to realistically alter specific regions — replacing a genuine expiry date with a manipulated one, or inserting realistic-looking damage or mold into a fresh product photo. This module specifically targets these localized GAN-based edits.

GAN Artifacts Detected

- Spectral replication artifacts: GAN upsampling components introduce characteristic frequency-domain artifacts that appear as replications in the Fourier spectrum.
- Checkerboard artifacts: Transposed convolution upsampling in GAN generators produces characteristic checkerboard patterns in synthesized regions, visible in frequency analysis.

- Color channel statistical discrepancies: GAN-generated regions often have subtle color channel correlations that differ from surrounding genuine image regions.
- Boundary blending inconsistencies: Even sophisticated GAN inpainting leaves statistical traces at the boundary between genuine and generated regions.
- Texture uniformity anomalies: GAN-generated text and numbers on product packaging show characteristic smoothness that differs from the natural variation of physically printed text.

Detection Approach

VeriSight implements AutoGAN-style spectrum-based classification (IEEE 2019) combined with GAN re-synthesis detection — comparing the input image with its re-synthesized version across super-resolution, denoising, and colorization tasks. Discrepancies between the original and re-synthesized versions reveal GAN-inserted regions.

4.4 Module 4 — OCR-Assisted Text & Expiry Date Verification (Weight: 15%)

Purpose

This is VeriSight's most distinctive module — one that no existing general-purpose image forensics system implements for the refund fraud use case. It uses OCR to extract and analyze text content from product packaging, specifically to detect digitally altered expiry dates, batch numbers, and label text.

> **Critical Design Principle**
> OCR plays a supporting role in VeriSight — it assists verification but is not used alone for decision-making. It is combined with image forensics evidence from Modules 1–3 before contributing to the final Authenticity Confidence Score.

OCR Pipeline Steps

1. Image preprocessing: Grayscale conversion, CLAHE contrast enhancement, adaptive thresholding, noise reduction to maximize OCR accuracy on varied packaging surfaces.
2. Text region detection: YOLOv8 or DBNet+CBAM localizes all text regions including dot-matrix printed expiry dates, laser-etched codes, and standard label text.
3. Text recognition: EasyOCR (CRNN + CRAFT architecture) or PaddleOCR extracts text content; handles dot-matrix fonts, varied date formats (DD/MM/YY, MM-YYYY, DD.MM.YYYY), and multiple scripts.
4. Date extraction & validation: Regex pattern matching extracts all date-format strings and identifies the most likely expiry date candidate from the recognized text.
5. Statistical plausibility analysis: The extracted date is analyzed for statistical plausibility against the platform's order date and delivery timestamp.
6. Text texture forensics: The visual texture of the extracted text region is analyzed by the CNN to detect whether the text was digitally inserted vs. physically printed.

5. Authenticity Confidence Score Engine

The outputs of all four modules are synthesized by the Authenticity Confidence Score Engine into a single score from 0 to 100, where 100 means the image appears completely genuine and 0 means it is almost certainly manipulated.

Module	Weight	Signal Provided
CNN Image Forensics	**35%**	Forgery probability: ELA map, copy-move score, splicing score, boundary analysis.
Transformer AI Detection	**30%**	AI-generation probability: ViT attention score, CLIP embedding distance, frequency spectrum.
GAN Artifact Analysis	**20%**	GAN manipulation probability: spectral replication, texture anomaly, boundary blending score.
OCR Text Verification	**15%**	Text plausibility: date validity, font texture forensics, text region CNN analysis.

5.1 Decision Thresholds

Score	Classification	Platform Action	Notes
85–100	**Genuine**	Auto-approve refund	Verified authentic; processed within 60 seconds.
60–84	**Likely Genuine**	Fast-track review	High probability genuine; human reviewer spot-checks within 2 hours.
35–59	**Suspicious**	Manual review + proof request	Refund held; customer asked for additional video or original packaging photo.
0–34	**Manipulated**	Reject + flag account	Refund denied; account flagged; evidence report generated.

5.2 Explainable AI (XAI)

Every VeriSight decision is fully explainable. The system generates a human-readable evidence report for each complaint evaluation containing:

- Overall Authenticity Confidence Score with module-level breakdown.
- Visual heatmap overlay on the complaint image showing suspected manipulation regions (GCA-Net + Grad-CAM).
- SHAP-based feature importance explanation showing which signals drove the score.
- OCR-extracted text with date plausibility analysis result.
- Recommended action with confidence level.

This explainability is critical — it enables human reviewers to audit every decision, understand edge cases, and override the system when needed. It also makes VeriSight outputs usable as legal evidence.

6. System Architecture

VeriSight is built as a cloud-ready, API-first microservice system designed to integrate directly into the existing complaint and refund workflows of any delivery platform via REST API, with a target end-to-end processing latency of under 5 seconds per image.

6.1 Processing Pipeline

7. Customer submits complaint with product image via platform app.
8. Platform sends image + complaint metadata to VeriSight REST API.
9. API Gateway validates the request and queues the image for processing.
10. Image is routed simultaneously to all four analysis modules (parallel processing via Celery).
11. All four modules return results: forgery probability, AI-generation probability, GAN artifact score, OCR text plausibility.
12. Score Engine aggregates all four outputs into Authenticity Confidence Score (0–100).
13. SHAP + Grad-CAM generate explainability report with visual heatmap overlay.
14. Final response returned to platform API: score + decision + explanation + heatmap image.
15. Platform updates complaint status and takes appropriate action.

6.2 Architecture Layers

Layer	Component	Technology
API Gateway	Complaint intake endpoint	FastAPI (async) + Pydantic validation + JWT authentication
Parallel Processing	Module orchestration	Celery distributed task queue + Redis broker for concurrent module execution
CNN Module	Image forensics inference	PyTorch + EfficientNet-B4 + ONNX Runtime + OpenCV for ELA
Transformer Module	AI-image detection inference	PyTorch + ViT-B/16 + HuggingFace Transformers + CLIP (OpenAI)
GAN Module	GAN artifact analysis	NumPy + SciPy (FFT spectrum analysis) + AutoGAN spectrum classifier
OCR Module	Text extraction pipeline	EasyOCR + PaddleOCR + YOLOv8 text detection + Regex date parsing
Score Engine	Score aggregation + XAI	Python + SHAP + Grad-CAM visualization + PIL for heatmap overlay
Database	Complaint & result storage	PostgreSQL (complaint history, audit log) + Redis (feature cache, model cache)
Frontend Dashboard	Admin monitoring UI	React.js + Recharts (analytics) + Leaflet.js (geographic visualization)
Infrastructure	Cloud deployment	Docker + Kubernetes + AWS EC2 / GCP + GitHub Actions (CI/CD)

7. Full Technology Stack

7.1 AI & ML Stack

Technology	Purpose & Justification
PyTorch	Primary deep learning framework for training and serving all CNN, ViT, and GAN detection models. Dynamic computation graph enables flexible model architecture experimentation.
EfficientNet-B4	CNN backbone for image forensics — state-of-the-art accuracy-to-parameter ratio. Pre-trained on ImageNet, fine-tuned on CASIA 2.0 forgery dataset.
Vision Transformer (ViT-B/16)	AI-generated image detector — self-attention mechanisms detect cross-patch semantic inconsistencies; achieves 96% accuracy on synthetic image detection tasks.
CLIP (OpenAI)	Vision-language model for GAN fingerprinting — CLIP-ResNet50 achieves 99.4% accuracy on StyleGAN/StyleGAN2 detection (CVPR 2023).
GCA-Net	Pixel-level forgery localization using gated context attention — produces interpretable manipulation heatmaps for human reviewer display.
EasyOCR + PaddleOCR	Text extraction from product packaging — EasyOCR handles varied fonts and layouts; PaddleOCR handles dot-matrix expiry date printing.
YOLOv8 / DBNet+CBAM	Text region detection on product packaging — localizes expiry date fields, label text, and barcode regions for targeted OCR extraction.
SHAP + Grad-CAM	Explainable AI — SHAP provides feature-level score attribution; Grad-CAM generates visual activation maps showing which image regions drove the model decision.
ONNX Runtime	Optimized model inference runtime for production deployment — converts PyTorch models to ONNX for platform-independent, low-latency inference.
OpenCV + Pillow	Image preprocessing (grayscale, thresholding, noise reduction), ELA map generation, heatmap overlay rendering on complaint images.

7.2 Backend & Infrastructure Stack

Technology	Purpose & Justification
Python 3.11	Primary backend language — extensive ML/AI library ecosystem makes Python the only practical choice for this system.
FastAPI	High-performance async REST API framework — async request handling enables parallel module processing without blocking; automatic OpenAPI documentation for easy platform integration.
Celery + Redis	Distributed async task queue — enables all four VeriSight modules to run in parallel rather than sequentially, achieving sub-5-second end-to-end latency.

PostgreSQL	Primary relational database for storing complaint records, VeriSight score history, fraud flags, account-level analytics, and full audit trail.
React.js	Admin monitoring dashboard — real-time fraud score feeds, complaint review queue, analytics charts, and heatmap visualization.
Docker + Kubernetes	Containerized deployment for consistent, scalable, cloud-portable infrastructure. GitHub Actions for CI/CD automation.

8. Dataset Strategy

VeriSight requires training data across two distinct domains: general image forgery detection (using existing public benchmarks) and product-specific packaging/expiry date manipulation (custom dataset to be built). The team has a concrete three-pronged strategy.

8.1 Public Benchmark Datasets

Dataset	Size	Use in VeriSight
CASIA 2.0	12,614 images	Primary CNN training dataset for splicing and copy-move forgery detection.
CoMoFoD	260 image sets	Copy-move forgery with post-processing robustness testing.
DF2023	1M+ images	Large-scale training for model generalization; covers diverse forgery types.
MICC-F220	220 images	Pixel-level forgery localization ground truth for GCA-Net heatmap validation.
ImagiNet	200,000 images	Real vs AI-generated image classification training for ViT module.
FaceForensics++	1,000 videos	Deepfake / GAN detection pre-training for the AI-image detection module.

8.2 Custom VeriSight Product Dataset

Since no public dataset exists specifically for product packaging image manipulation in the refund fraud context, the team constructs a custom dataset targeted at 4,000+ labeled product complaint images across 5 classes: genuine, expiry-manipulated, label-manipulated, AI-generated, and GAN-inpainted.

16. Photograph 150+ real product packages from categories: snacks, beverages, dairy, medicines, cosmetics, grocery.
17. Create genuine complaint versions: photograph packages at slightly poor angles and under varying lighting conditions.
18. Create expiry-date-manipulated versions: digitally alter dates using Photoshop, Canva, and AI inpainting tools.
19. Create label-manipulated versions: alter brand names, weights, MRPs, and batch numbers on packaging.
20. Create AI-generated versions: generate synthetic product complaint images using Stable Diffusion, DALL-E 3, and Adobe Firefly.
21. Apply augmentation: rotation, brightness variation, JPEG re-compression at varied quality levels, and noise addition.
22. Label dataset: manipulation type, manipulated region pixel mask, manipulation tool used, confidence level.

9. Model Evaluation Metrics & Targets

Metric	Target	Context
CNN Accuracy	**> 93%**	Baseline: ResNet-101 at 93.46% on CASIA 2.0 (IEEE 2024). VeriSight targets 95%+ using EfficientNet-B4 + ELA.
ViT AI-Image Accuracy	**> 95%**	ViTs achieve up to 96% accuracy on AI-generated image detection tasks.
CLIP GAN Accuracy	**> 99%**	CLIP-ResNet50: 99.4% on StyleGAN, 99.2% on StyleGAN2 (CVPR 2023).
OCR Date Extraction F1	**> 88%**	R-CNN achieves 91.1% precision, 80.9% F1 on pharmaceutical packaging text (SCPE 2024).
Overall System Precision	**> 91%**	High precision protects genuine customers from unjust refund rejections.
Overall System Recall	**> 89%**	High recall ensures manipulated images are not approved as genuine.
End-to-End Latency	**< 5 seconds**	Parallel module execution via Celery targets sub-5-second total response time.

10. Hackathon Development Roadmap

Phase 1 — Core Model Training (Hours 1–6)

- Download CASIA 2.0 and set up EfficientNet-B4 training pipeline.
- Implement ELA map generation and train CNN-ELA combined model.
- Load pre-trained ViT-B/16 and CLIP-ResNet50 for AI-image detection.
- Implement AutoGAN-style spectrum classifier for GAN artifact detection.
- Set up EasyOCR pipeline with YOLOv8 text region detection.

Phase 2 — Integration & API (Hours 6–14)

- Build FastAPI backend with complaint image intake endpoint.
- Implement Celery parallel task queue for concurrent module execution.
- Build Authenticity Confidence Score aggregation engine.
- Integrate SHAP + Grad-CAM for explainability report generation.
- Set up PostgreSQL schema for complaint tracking.

Phase 3 — Dashboard & Custom Dataset (Hours 14–20)

- Build React.js admin dashboard with real-time score visualization.
- Photograph and prepare 200+ product images for custom dataset.
- Create manipulated versions (expiry date, label edits) and label dataset.
- Fine-tune OCR pipeline on custom product packaging images.
- Implement heatmap overlay visualization on complaint images.

Phase 4 — Demo Preparation (Hours 20–24)

- End-to-end testing: submit genuine images vs manipulated images and verify scores.
- Prepare live demo: expiry date manipulation detection in real time.
- Performance benchmark: accuracy, precision, recall, F1, latency.
- Record backup demo video in case of live demo issues.
- Prepare pitch and system walkthrough for judges.

11. Unique & Differentiating Features

Feature	Description
Expiry Date Fraud Focus	VeriSight is the only system that specifically targets digitally altered expiry dates on product packaging — the most common manipulation vector in quick commerce and grocery delivery fraud.
Four-Layer Parallel Analysis	CNN forensics + Transformer AI detection + GAN artifact analysis + OCR verification all run simultaneously — no single point of failure; each layer catches different fraud types.
Platform-Agnostic API	REST API design means VeriSight can be integrated into any delivery, e-commerce, or quick commerce platform in under a day — Zomato, Swiggy, Blinkit, BigBasket, Amazon Fresh, Flipkart.
Visual Heatmap Evidence	Generates Grad-CAM + GCA-Net heatmap overlays showing exactly which regions were identified as manipulated — makes decisions auditable and legally defensible.
Explainable Score Report	Every Authenticity Confidence Score comes with a SHAP-based breakdown explaining which of the four modules flagged the image and why — human reviewers can understand and override every decision.
Production-Ready Design	Architecture choices (FastAPI, Docker, ONNX, REST APIs) are all production-deployment decisions — VeriSight is designed to be actually integrated into real systems.
OCR as Supporting Evidence	OCR-extracted text plausibility analysis is used to support (not replace) image forensics decisions — this nuanced design prevents OCR failures from causing wrong decisions.

12. Elevator Pitch

VeriSight in 60 Seconds

Every day, delivery platforms blindly trust customer-uploaded images to approve refunds — and fraudsters know it. They alter expiry dates, manipulate labels, and use AI tools to generate fake complaint photos that look completely real. VeriSight stops them.

Our AI-powered visual verification system analyzes every complaint image across four layers — image forensics (CNN), AI-generation detection (ViT + CLIP), GAN artifact analysis, and OCR text verification — producing an Authenticity Confidence Score in under 5 seconds.

No manual reviewer needed for the clear cases. Every decision is explainable. Every manipulation is visualized. VeriSight does not just detect fraud — it makes platforms smarter, fairer, and more trustworthy for everyone.

Why VeriSight Should Win FORGE-X 2026

Real-World Problem at Massive Scale

Returns fraud cost merchants USD 103 billion in 2023. In India, 45% of consumers have encountered deepfake shopping scams. This is not a hypothetical future problem — it is happening right now, at scale, on every platform the judges use.

Addresses a Gap No One Has Solved

No automated system specifically verifies the authenticity of user-submitted product complaint images. Every platform relies on human reviewers. VeriSight solves a real, current, unaddressed gap in the industry's fraud defense stack.

Grounded in Rigorous Research

25+ peer-reviewed papers from IEEE, Springer, NeurIPS, CVPR, MDPI, and ACM directly support each technical decision in VeriSight. This is not AI buzzword stacking — every module has a research-validated foundation.

Practical, Integrable, and Deployable

VeriSight is built with production deployment as the primary design constraint. REST APIs, Docker containers, ONNX inference, async processing — every technical choice reflects a system meant to be used, not just demoed.

Explainable, Auditable, and Fair

VeriSight does not black-box fraud decisions. Every Authenticity Confidence Score comes with a full explanation. Genuine customers who are incorrectly flagged can be reviewed transparently. This is ethical AI design in practice.

— End of VeriSight Project Document — FORGE-X 2026 — Team: return0 —